

LEGIRA: An AI-Assisted Legal Case Prioritization Framework for Fairer Undertrial Review

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Abstract

Judicial backlogs produce both operational inefficiency and fairness risk. Cases involving prolonged undertrial detention, repeated adjournments, and vulnerable litigants require earlier review than ordinary queue-based handling can provide. LEGIRA is an AI-assisted legal triage framework developed during the hackathon to identify such matters and prioritize them through a legally grounded scoring pipeline. The system integrates structured case datasets, statute-derived sentence estimates, vulnerability indicators, temporal delay signals, and lightweight NLP-derived features to rank matters and recommend operational tracks such as Critical-Review, Bail-Eligible, and Fast-Track. The implementation combines explicit fairness rules with machine learning models and clustering-based backlog segmentation and exposes the outputs through a Streamlit dashboard. Based on the checked-in evaluation artifacts, the system achieved 88.43 percent urgency classification accuracy, 73.97 percent case-type classification accuracy, 100 percent recall for undertrial overstay detection, and a dashboard filter response time of approximately 3.26 milliseconds.

Keywords: Legal AI, judicial backlog, undertrial detection, case prioritization, explainable machine learning, fairness-aware triage

I. INTRODUCTION

Large judicial systems typically manage pending cases through administrative queues, but such queues do not sufficiently

distinguish between ordinary delay and fairness-critical delay. LEGIRA was built to support courts and reviewers by ranking cases based on detention history, legal severity, vulnerability, and procedural delay rather than relying only on filing order.

II. PROBLEM STATEMENT AND OBJECTIVE

A. Problem Statement

Court backlogs create operational and fairness risks. Undertrial prisoners may remain in custody longer than justified, vulnerable persons may not receive timely review, repeated adjournments may accumulate unnoticed, and manual queues can become inconsistent and difficult to audit.

B. Objective

The objective of LEGIRA is to identify high-risk and unfairly delayed cases, detect undertrial overstay with very high recall, produce an interpretable priority score, recommend handling tracks such as Critical-Review and Bail-Eligible, and evaluate baseline and improved machine learning approaches.

III. METHODOLOGY / APPROACH

A. Data Integration

The pipeline merges five structured datasets: core case metadata, detention records, temporal hearing history, demographic and vulnerability indicators, and NLP-derived case signals. A statute lookup table maps

IPC and BNS sections to punishment and bailability context. The repository reflects two implementation stages: Phase 1 baseline workflow and Phase 2 extended comparison workflow.

B. Feature Engineering

The system computes undertrial status, final sentence estimates, detention ratio, overstay flags, vulnerability component, stale case indicators, offense priority component, and the final priority score. Additional technical signals include statutory maximum sentence days, non-finite sentence flags, legal bailability flags, age-vulnerable, gender-vulnerable, health-vulnerable, detention ratio exponential normalization, and offense severity index. The baseline score uses weighted contributions from overstay, detention ratio, offense priority, vulnerability, and stale hearing history, followed by a critical override for undertrial overstay.

C. Modeling Strategy

The Similar Model applies Random Forest classification with KMeans clustering. The Improved Model applies XGBoost or a gradient-boosting fallback with agglomerative hierarchical clustering and cross-validation support. Both predict urgency level, recommended track, and backlog segment.

D. Output Workflow

The system produces a ranked hearing queue, flagged undertrial list, synthetic demo case table, cluster summary, evaluation metrics, model comparison artifacts, and serialized model files.

IV. SYSTEM ARCHITECTURE

A. Phase 1 Architecture

Phase 1 is the baseline decision-support architecture. Its workflow is: raw CSV datasets to load and merge, statute lookup, feature engineering, priority score generation, Random Forest urgency and track models, KMeans clustering, ranked and flagged CSV outputs, and dashboard visualization.

Phase 1 file architecture from D:- 1 excluding hidden git content is: README.md, dashboard.py, prioritization_engine.py, generate_statute_lookup.py, requirements.txt, the data folder with the five datasets plus statute files, and the outputs folder with cluster summary, evaluation metrics, flagged undertrials, prioritized cases, and synthetic demo cases.

B. Phase 2 Architecture

Phase 2 extends the system into an evaluation architecture. Its workflow is: raw legal datasets to prioritization_engine.py, feature engineering and baseline scoring, SimilarModel using Random Forest plus KMeans, ImprovedModel using XGBoost or boosting plus hierarchical clustering, compare_models.py, model comparison outputs, sample predictions, saved pickle artifacts, and enhanced dashboard review.

Phase 2 file architecture from D:- 2 and 3 excluding hidden git content includes logo and package files, the hackathon2 folder with compare_models.py, dashboard.py, generate_statute_lookup.py, model_improved.py, model_similar.py, prioritization_engine.py, requirements.txt, data, outputs, and comparison_outputs, plus the js-crypto-model1 folder containing the blockchain prototype files. For readability, generated and vendor-heavy folders such as

virtual environments, `node_modules`, and `__pycache__` are omitted from the authored architecture summary.

V. IMPLEMENTATION DETAILS

A. Technology Stack

The legal analytics portion is implemented in Python using pandas, numpy, scikit-learn, streamlit, and altair. The improved model additionally supports xgboost, lightgbm, and imbalanced-learn. A separate exploratory audit prototype is implemented in Node.js using express, ws, crypto-js, elliptic, and jest.

B. Pipeline Stages

The Python pipeline is organized into data loading, legal normalization, feature derivation, supervised classification, unsupervised segmentation, and artifact generation. Section parsing is handled using regular-expression tokenization, while sentence ceilings are inferred from statutory lookup tables.

C. Priority Logic

The priority score uses weighted contributions from overstay status, detention ratio, offense and bail sensitivity, vulnerability, and stale hearing history. Cases are routed into Critical-Review, Bail-Eligible, Fast-Track, Regular-Criminal, or Civil tracks. The implementation also emits human-readable flag reasons such as undertrial overstay beyond statutory or derived sentence limit, stale hearing history, and health or disability vulnerability.

D. Dashboard Support

The Streamlit dashboard supports ranked

case review, flagged undertrial inspection, cluster-level backlog viewing, new-case scoring, and text-assisted summarization hooks. The enhanced dashboard also supports local LLM-assisted case summarization.

VI. RESULTS / OUTPUTS

A. Baseline Evaluation Metrics

From the checked-in evaluation output, the system achieved 88.43 percent urgency classification accuracy, 73.97 percent case-type classification accuracy, 100 percent undertrial detection recall, 3.26 millisecond dashboard filter response time, and a ranked-list agreement of 0.8942 by Spearman correlation.

B. Model Comparison

The Similar Model achieved 88.43 percent urgency accuracy and 68.78 percent weighted track F1, while the Improved Model achieved 88.84 percent urgency accuracy and 70.39 percent weighted track F1. Both models preserved 100 percent undertrial recall.

C. Generated Output Scale

The checked-in outputs include 1,206 ranked cases, 321 flagged undertrial cases, 6 synthetic demo cases, and 4 cluster groups. The highest-risk segment, Cluster-A Critical Backlog, has a mean priority score above 92.

D. Practical Output Interpretation

The ranked records are accompanied by interpretable reasons such as undertrial overstay beyond statutory or derived sentence limit, stale hearing history, and

health or disability vulnerability.

Legis

ADMIN VIEW

NAVIGATION

Dashboard

All Cases

Priority Queue

Register Case

Blockchain

ACCOUNT

Sign Out

System Admin

admin@legis.gov.in

Priority Queue

Fri, 17 Apr, 2026

16 CRITICAL16 HIGH56 MEDIUM12 LOW100 Total

Search...

Refresh

RANK	CASE #	TYPE	OFFENSE	JUDGE	PRIORITY	SCORE	OVERSTAY	DETENTION	PENDING	STATUS	ACTIONS
1	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→
2	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→
3	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→
4	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→
5	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→
6	#6	Criminal	Domestic Violence	Justice Ananya Singh	CRITICAL	90.51 /100	OVERSTAY	463d	463d	Active	→

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Case Details

Fri, 17 Apr, 2026

Case #6ActiveCRITICAL

Criminal · Domestic Violence · District Court Bangalore · Justice Ananya Singh

Priority Score: 90.51/100

Update StatusSchedule HearingRecord HearingAssign Advocate

Deliver Judgment

OverviewPersons (3)Advocates (2)Hearings (1)TimelineStatus Verification

Case Details

Case ID#6

Case TypeCriminal

Offense TypeDomestic Violence

Legal Statusunder_trial

Trial Statusin_progress

Current Stagearguments

Filing Date2024-01-08

Last Hearing2026-04-11

Registered ByMohan Lal Gupta

CourtDistrict Court Bangalore

JudgeJustice Ananya Singh

Created At2026-04-16

FIR Text

Detention Info

Detention Days

Expected Sentence

Detention Ratio

Overstay

Life Sentence

Trial Data

Days Pending

Number of Trials

Adjournments

Last Hearing Gap

Admin Assistant

Powered by Qwen2.5 · Personalized for System

Hello, System!

I have access to your system data & analytics. Ask me anything.

How many total cases are in the system?

Which cases have overstay alerts?

How many users are registered?

Show me today's system statistics.

Ask about your cases, hearings, priorities...

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ADMIN VIEW

NAVIGATION

Dashboard

All Cases

Priority Queue

Register Case

Blockchain

ACCOUNT

Sign Out

System Admin

admin@legira.gov.in

Legira

Fri, 17 Apr, 2026

Administrator

Blockchain Ledger

Immutable audit trail for all case lifecycle actions

Verify Integrity

50

Blocks

49

Transactions

12

Cases Tracked

NO

Chain Valid

#49

Last Block

TAMPERING DETECTED

49 block(s) have been altered. Chain integrity compromised.

Blocks (50)

Audit Trail (49)

Chain Visual

Analytics

#	HASH	PREV HASH	TRANSACTIONS	NONCE	DIFF	MINED AT
#49	000b370e50719848...	00c0f1a79897188e...	1	506	3	17 Apr 2026, 06:42:22 am
#48	00c0f1a79897188e...	0983effcaa3bc0bd...	1	7	2	17 Apr 2026, 06:42:21 am
#47	0983effcaa3bc0bd...	08b639df706535dd...	1	4	1	17 Apr 2026, 06:42:21 am
#46	08b639df706535dd...	00c9408174db8c2b...	1	2	1	17 Apr 2026, 06:42:20 am

Legira

WRITER VIEW

NAVIGATION

Dashboard

Register Case

My Cases

Blockchain

ACCOUNT

Sign Out

Clerk Priya Sharma

writer@legira.gov.in

Legira

Fri, 17 Apr, 2026

Writer/Clerk

Blockchain Ledger

Immutable audit trail for all case lifecycle actions

Verify Integrity

1

Blocks

0

Transactions

0

Cases Tracked

—

Chain Valid

#0

Last Block

Blocks (1)

Audit Trail (0)

Chain Visual

#	HASH	PREV HASH	TRANSACTIONS	NONCE	DIFF	MINED AT
#0	05e2d5a587998990...	0000000000000000...	Genesis	0	3	17 Apr 2026, 09:40:41 am

<<

<

1

>

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WRITER VIEW

NAVIGATION

Dashboard

Register Case

My Cases

Blockchain

ACCOUNT

Sign Out

Clerk Priya Sharma

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9. Action Taken

The case has been registered under relevant sections of IPC. Investigation has been initiated. The crime scene has been secured, and forensic teams have been dispatched. The body has been sent for post-mortem examination.

10. Signature

Complainant Signature: Ramesh Kumar

Officer-in-Charge:
Inspector Rajesh
Central Crime Police Station

End of FIR

CASE SUMMARY

Generate Summary from FIR

The case involves the brutal murder of Arjun Prakash, a businessman, who was found with multiple stab wounds and signs of blunt force trauma near an abandoned warehouse on Old Mahabalipuram Road in Chennai. The incident occurred between 10:30 PM and 11:30 PM on April 16, 2026. The crime is registered under Sections 302 IPC (Murder), 120B IPC (Criminal Conspiracy), and 201 IPC (Destruction of Evidence). The complainant, Ramesh Kumar, reported the case to the Central Crime Police Station on April 17, 2026.

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Legal Sections / Charges

AI Suggested Sections

Accept All

Dismiss All

IPC §302 — Murder

Brutal Murder

✓

✗

IPC §120B — Criminal conspiracy

Criminal Conspiracy

✓

✗

IPC §201 — Causing disappearance of evidence of offence

Destruction of Evidence

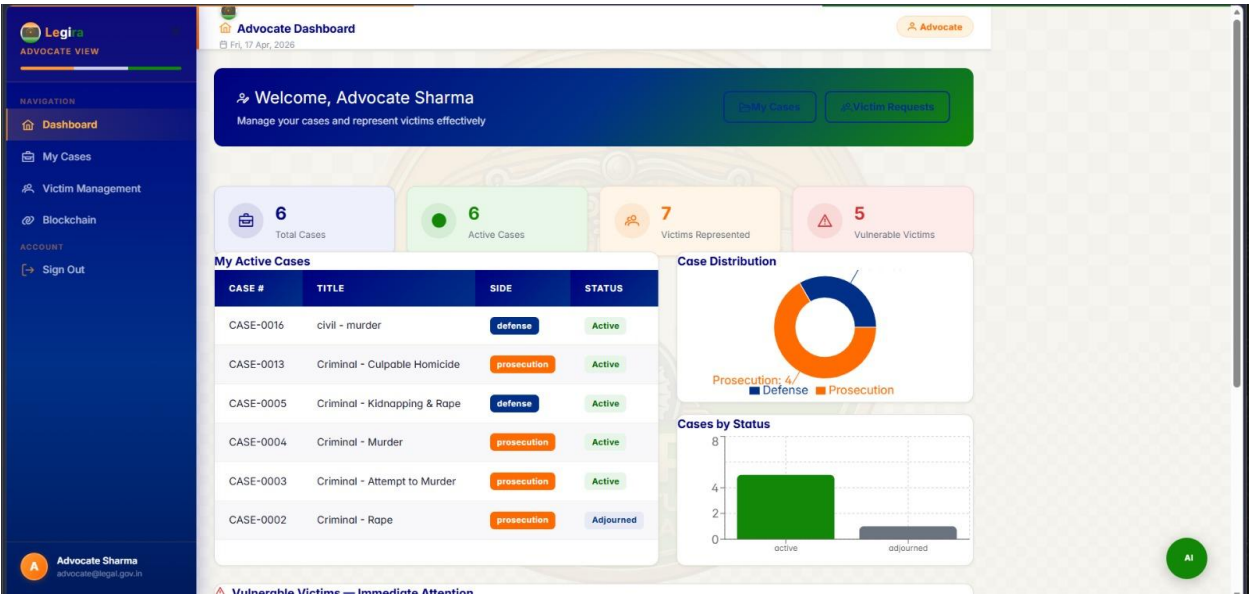
✓

✗

Search section (e.g. 302, IPC, Murder)

SELECT	ACT	SECTION	DESCRIPTION	BAIL	MAX YRS
+	IPC	120B	Criminal conspiracy	Non-Bailable	2
+	IPC	201	Causing disappearance of evidence of offence	Non-Bailable	7
+	IPC	302	Murder	Non-Bailable	99
+	IPC	304	Culpable homicide not amounting to murder	Non-Bailable	10

AI suggests 3 applicable IPC section(s) based on case classification.



VII. CONCLUSION

LEGIRA demonstrates that fairness-aware legal triage can be implemented as an explainable AI workflow rather than a black-box prediction system. By combining statutory sentence references, detention measures, vulnerability indicators, and delay signals, the system creates a ranked hearing queue that is more aligned with urgency and justice concerns than simple filing-order review. Its strongest result is 100 percent checked-in recall for undertrial overstay detection, supported by strong urgency-classification performance and fast dashboard filtering.

VIII. FUTURE SCOPE

Future work should focus on integration with real court information systems, stronger legal text understanding, reviewer override logging, immutable audit trails, fairness audits with domain experts, and production-grade packaging and testing.

REFERENCES

[1] Reference/methodology.txt

[2] Phases/Phase - 1/prioritization_engine.py

[3] Phases/Phase - 1/dashboard.py

[4] Phases/Phase - 2
3/hackathon2/prioritization_engine.py

[5] Phases/Phase - 2
3/hackathon2/model_similar.py

[6] Phases/Phase - 2
3/hackathon2/model_improved.py

[7] Phases/Phase - 2
3/hackathon2/compare_models.py

[8] Phases/Phase - 2
3/hackathon2/outputs/evaluation_metrics.js
on

[9] Phases/Phase - 2
3/hackathon2/comparison_outputs/model_co
mparison.csv

[10] Phases/Phase - 2
3/hackathon2/comparison_outputs/detailed_

metrics.json